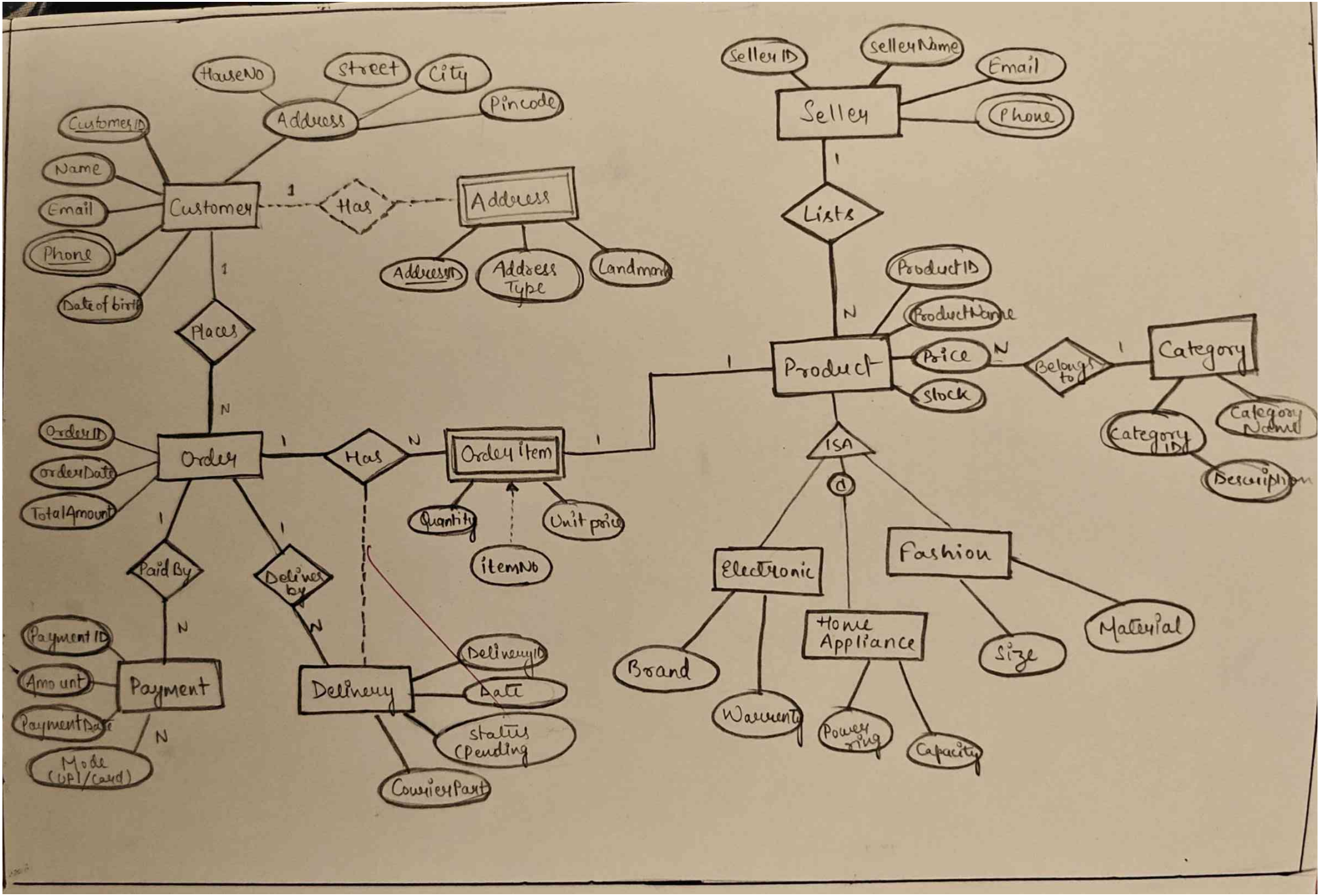

HouseNo
Street
City
Pincode
Address
CustomerID
Name
Email
Phone
Date of birth
Customer
1
Has
Address
AddressID
Address Type
Landmark
1
Places
N
SellerID
SellerName
Email
Phone
Seller
1
Lists
N
ProductID
ProductName
Price
N
Belongs to
1
Category
Product
1
Stock
CategoryID
CategoryName
Description
ISA
d
Electronic
Fashion
Home Appliance
Brand
Warranty
Power rating
Capacity
Size
Material
OrderID
orderDate
TotalAmount
Order
1
Has
N
Order item
1
Quantity
Unit price
itemNo
1
Paid By
1
Delivered by
N
N
PaymentID
Amount
PaymentDate
Mode (UPI/card)
Payment
N
DeliveryID
Date
status (Pending
CourierPart
Delivery

EXPERIMENT-01

AIM - To design an entity-Relationship (ER) diagram for an Indian E-commerce platform and Identify entities, attributes, relationships and cardinalities.

OBJECTIVES - To identify entities, attributes, relationship keys, weak entities, specialization, and participation constraints in an e-commerce database system.

ENTITIES - Customer
Address
Seller
Category
Product
Order
Payment
Delivery

CARDINALITY:
Customer & order = 1:N
ORDER & ORDER_ITEM = 1:N
CATEGORY & PRODUCT = 1:N
CUSTOMER & ADDRESS = 1:N
SELLER & PRODUCT = 1:N
Payment & order = 1:1
Address & order = 1:N

ER DIAGRAM NOTATION SUMMERY

- Entity
- Weak Entity
- Attribute
- Multi-valued Attribute
- Partial key
- Total participation
- Partial participation
- Relationship
- Identifying Relationship
- ISA (specialization)
- d Disjoint specialization
- I,N Cardinality

CONCLUSION

The ER Diagram consists of Customer, Product, seller, Category order, orderItem, Payment, Delivery, and Address entities. Address and orderItem is a weak entity, Product has specialization into Electronics, Fashion, and Grocery, and the system follow appriciate participation constraints and relationship.